

Variational Autoencoder

Assume data $X = (x_i)_{i=1}^N$ are i.i.d. with an unknown distribution. Assume the latent variable z has prior distribution $z \sim p_\theta(z)$ and the likelihood $p_\theta(x|z)$ is parametrized by learnable parameter θ .

The central problem is to determine the posterior $p_\theta(z|x)$.

Variational Autoencoder uses a parametrized family $q_\phi(z|x)$ to approximate $p_\theta(z|x)$.

We derive a proper objective function. We claim that for $x_i \in X$

$$\log p_\theta(x_i) = \text{KL}(q_\phi(z|x_i) \parallel p_\theta(z|x_i)) + \mathbb{E}_{z \sim q_\phi(z|x_i)} \left[\log \frac{p_\theta(x_i, z)}{q_\phi(z|x_i)} \right] \quad (1)$$

This is because

$$\text{KL}(q_\phi(z|x_i) \parallel p_\theta(z|x_i)) = \mathbb{E}_{z \sim q_\phi(z|x_i)} \left[\log \frac{q_\phi(z|x_i)}{p_\theta(z|x_i)} \right].$$

Hence RHS of (1) is

$$\mathbb{E}_{z \sim q_\phi(z|x_i)} \left[\log \frac{p_\theta(x_i, z)}{p_\theta(z|x_i)} \right] = \mathbb{E}_{z \sim q_\phi(z|x_i)} [\log p_\theta(x_i)] = \log p_\theta(x_i).$$

We denote

$$L(\theta, \phi, x_i) := \mathbb{E}_{z \sim q_\phi(z|x_i)} \left[\log \frac{p_\theta(x_i, z)}{q_\phi(z|x_i)} \right]$$

and it is the **evidence lower bound** (ELBO).

Because of (1), maximizing $L(\theta, \phi, x_i)$ with respect to θ and ϕ will maximize $p_\theta(x_i)$ and minimize the KL divergence between the approximated and true posteriors.

We next derive another form of ELBO that may be more suitable for practical

computation.

Note that the ELBO

$$\begin{aligned}
L(\theta, \phi, x_i) &= \mathbb{E}_{z \sim q_\phi(z|x_i)} [\log p_\theta(x_i, z) - \log q_\phi(z|x_i)] \quad (*) \\
&= \mathbb{E}_{z \sim q_\phi(z|x_i)} [\log p_\theta(z) - \log q_\phi(z|x_i) + \log p_\theta(x_i, z) - \log p_\theta(z)] \\
&= -\text{KL}(q_\phi(z|x_i) \parallel p_\theta(z)) + \mathbb{E}_{z \sim q_\phi(z|x_i)} [\log p_\theta(x_i|z)] \quad (**)
\end{aligned}$$

The way to compute the expectation is to use the Monte Carlo method:

$$\mathbb{E}_{z \sim p(z)} [f(z)] \approx \frac{1}{L} \sum_{l=1}^L f(z_l),$$

where z_l are sampled from $z_l \sim p(z)$.

To compute the derivative of L w.r.t θ , due to the Dominant Convergence Theorem, we can interchange the derivative and expectation and still use Monte Carlo:

$$\frac{\partial}{\partial \theta} L(\theta, \phi, x_i) = \mathbb{E}_{z \sim q_\phi(z|x_i)} \left[\frac{\partial}{\partial \theta} \log p_\theta(x_i, z) \right]$$

The derivative of L w.r.t ϕ is problematic, because ϕ appears in the distribution of the expectation. We may still switch the integral and derivative, but we may not be able to write the integral of the derivative as an expectation, and therefore, may not be able to use the Monte Carlo method.

A solution is to use **reparametrization**: Reparametrize $z \sim q_\phi(z|x)$ by $z = g_\phi(\varepsilon, x)$ with $\varepsilon \sim p(\varepsilon)$ for some function g_ϕ and distribution $p(\varepsilon)$. In this way ϕ is pulled out of the randomness. We have

$$\mathbb{E}_{z \sim q_\phi(z|x)} [f(z)] = \mathbb{E}_{\varepsilon \sim p(\varepsilon)} [f(g_\phi(\varepsilon, x))]$$

and the derivative w.r.t ϕ goes through the expectation.

Here are two examples for choices of reparametrization. One example is to assume $q_\phi(z|x)$ has tractable inverse CDF. Let $F_\phi(\varepsilon, x)$ be the (generalized) inverse of the CDF of $q_\phi(z|x)$. Then $F_\phi(\varepsilon, x)$ for ε uniform on $[0, 1]$ has the same distribution as $q_\phi(z|x)$. Another example is $z \sim q_\phi(z|x)$ and we reparametrize by $z = \mu_\phi(x) + \sigma_\phi(x)\varepsilon$ for ε being standard Gaussian.

Using reparametrization and Monte Carlo, two estimators of the ELBO L are offered. Estimator A, L^A , is based on (*).

$$L^A(\theta, \phi, x_i) = -\frac{1}{L} \sum_{l=1}^L \log q_\phi(z_{i,l}|x_i) + \frac{1}{L} \sum_{l=1}^L \log p_\theta(x_i, z_{i,l}),$$

where $z_{i,l} = g_\phi(\varepsilon_{i,l}, x_i)$ (reparametrization) and $\varepsilon_{i,l}$ are sampled from $p(\varepsilon)$. In case the KL divergence in (**) can be computed analytically,

$$L^B(\theta, \phi, x_i) = -\text{KL}(q_\phi(z|x_i) \parallel p_\theta(z)) + \frac{1}{L} \sum_{l=1}^L \log p_\theta(x_i|z_{i,l}),$$

where $z_{i,l} = g_\phi(\varepsilon_{i,l}, x_i)$ and $\varepsilon_{i,l} \sim p(\varepsilon)$ as above.

We give a concrete example. The MNIST dataset contains pictures of hand-written numbers from 0 to 9. Each picture is black and white and has 28×28 pixels. We let a picture be denoted as $x \in [0, 1]^N$ where $N = 28 \times 28$.

We let the latent variable $z \in \mathbb{R}^N$ and the prior be $z \sim \mathcal{N}(0, I)$. The likelihood is Bernoulli:

$$\log p_\theta(x|z) = \sum_{i=1}^N x_i \log f_\theta(z)_i + (1 - x_i) \log(1 - f_\theta(z)_i),$$

where $f_\theta(z) \in (0, 1)^N$ is represented by a neural network.

We let $q_\phi(z|x) = \mathcal{N}(z; \mu_\phi, (\sigma_\phi)^2 I)$ and reparametrize naturally by

$$z = \mu_\phi + \sigma_\phi \odot \varepsilon,$$

where $\odot$ is element-wise product and $\varepsilon \sim \mathcal{N}(0, I)$. The mean and standard deviation, $\mu_\phi = \mu_\phi(x)$ and $\sigma_\phi = \sigma_\phi(x)$, are represented by neural networks. We note that the KL divergence in this example, $\text{KL}(\mathcal{N}(z; \mu_\phi, (\sigma_\phi)^2 I) \parallel \mathcal{N}(0, I))$, is the KL

divergence of two Gaussian distributions, and can be computed analytically:

$$\begin{aligned}\text{KL}(\mathcal{N}(\mu_\phi, (\sigma_\phi)^2 I) \parallel \mathcal{N}(0, I)) &= -\frac{1}{2} \sum_{i=1}^N (1 + \log \sigma_i^2 - \mu_i^2 - \sigma_i^2) \\ &:= K(\mu_\phi, \sigma_\phi),\end{aligned}$$

where $\mu_\phi = (\mu_i)_i \in \mathbb{R}^N$ and $\sigma_\phi = (\sigma_i)_i \in \mathbb{R}^N$.

We therefore use type B estimator for the ELBO:

$$L(\theta, \phi, x) = -K(\mu_\phi, \sigma_\phi) + \frac{1}{L} \sum_{l=1}^L \log p_\theta(x \mid \mu_\phi + \sigma_\phi \odot \varepsilon_l)$$

and ε_l are sampled from $\varepsilon_l \sim \mathcal{N}(0, I)$.